

Electric Bus Fleet Operational Simulation for EV Charging Station Infrastructure Planning under Full TransJogja Electrification

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Abstract—Transit agencies planning full fleet electrification face two levers for keeping service regular: building more charging hardware, or scheduling the charging they already have more intelligently. Which lever matters more is rarely tested before the money is committed. We evaluate both on a full-network case study, the 20-corridor TransJogja bus rapid transit system in Yogyakarta, Indonesia, using a discrete-event simulation of fleet operations wrapped in a Monte Carlo layer ($N = 100$ days) that treats travel speed as log-normal with a 15% day-level and 5% segment-level coefficient of variation. Across 31 scenarios we find three results. Charging capacity behaves non-linearly: removing a single connector at 19 units raises degraded routes from 5.52 to 8.85 per day, a regime shift rather than a gradual decline. Hardware and scheduling then solve different problems. Raising capacity at the bottleneck station to 26 connectors gives the best headway (26.14 minutes) and places 17 of 20 routes within a ± 10 -minute tolerance, but leaves a residual degradation risk of 0.01 routes per day; a time-aware dynamic charging policy with no added hardware removes degradation entirely at a worse headway (33.10 minutes). Combining both holds zero degraded routes across a 14-day consecutive stress test at 30.78 minutes. No scenario meets the agency’s published headway targets on every route, which locates the remaining gap in the targets themselves rather than in the charging plan.

Index Terms—electric buses, discrete-event simulation, Monte Carlo simulation, charging infrastructure planning, bus rapid transit, capacity planning

I. INTRODUCTION

Electrifying a bus fleet is usually presented as a vehicle procurement problem. In practice the harder question is what happens to the timetable once the vehicles arrive. A diesel bus refuels in minutes and does so outside the service window; an electric bus that reaches a low state of charge mid-service must leave its route, travel to a charging station, queue behind other buses that made the same decision, and charge for tens of minutes before returning. Each of those steps consumes service time that the published schedule never budgeted for.

TransJogja, the bus rapid transit (BRT) operator in the Yogyakarta Special Region, Indonesia, illustrates the gap between a pilot and a full rollout. The operator currently runs two electric buses on a single corridor, charging at one public charging station in the Adisucipto area. The regional

transportation agency is considering extending electrification to all 20 corridors, which would put roughly 124 to 128 buses into service simultaneously. At that scale the pilot tells the agency almost nothing: two buses sharing one station never contend for a connector, while 128 buses drawing on a handful of stations contend constantly.

Testing the full configuration in the field is not an option. Installing charging hardware to find out whether the capacity was right costs on the order of the investment being questioned, and a failed configuration degrades service for actual passengers. Simulation is the standard substitute, but most published planning studies optimize a deterministic network model rather than simulating operations day by day, so they report an optimal configuration without reporting how often that configuration fails when traffic varies.

This paper reports an operational simulation study of the full electrification case. We model all 20 corridors running simultaneously as a discrete-event system, wrap it in a Monte Carlo layer so that each configuration is evaluated across 100 stochastic days rather than one average day, and compare 31 scenarios spanning station count, connector allocation, total capacity, and charging policy. Our contributions are:

- A full-network operational simulation of BRT electrification that reports headway degradation distributions rather than point estimates, validated qualitatively against the operator’s live single-corridor baseline.
- Evidence that charging capacity exhibits a sharp regime shift: a one-connector reduction near the threshold roughly doubles daily route degradation, which makes marginal capacity decisions near that point disproportionately consequential.
- A direct comparison showing that physical capacity and dynamic scheduling optimize different metrics, and that neither alone satisfies both reliability and headway targets.

II. RELATED WORK

Charging infrastructure planning for electric bus fleets is most often posed as a facility location and sizing problem.

Rogge et al. [1] jointly optimize fleet composition and charging infrastructure; Guschinsky et al. [2] and Liu et al. [3] treat station siting and capacity as mixed-integer programs. Zhou et al. [4], reviewing more than 100 studies of electric bus charging facility planning, report that the large majority adopt deterministic optimization formulations, and that only a small minority model uncertainty in travel time or energy consumption explicitly. Optimization answers where to build and how much, under the assumption that operations behave as the model's parameters say they will.

A second line of work simulates operations instead. Jefferies and Göhlich [5] build a discrete-event tool to size fleet, infrastructure, and energy demand on a real network, comparing depot and opportunity charging. Sebastiani et al. [6] apply a similar approach to BRT operations, confirming that discrete-event models represent queueing and resource contention at shared stations well. Casella et al. [7] extend the event-based view into a scheduling optimization with a non-linear charging profile. These studies establish that operational simulation captures the contention effects that matter here.

Energy consumption modelling forms the third input. Basma et al. [8] and Mamarikas et al. [9] show that consumption depends non-linearly on speed and driving dynamics, and Montanino et al. [10] demonstrate analytically and through large-scale Monte Carlo experiments that assuming constant speed systematically underestimates consumption, because energy use rises with acceleration variance.

The gap we address sits between these lines. Optimization studies report a configuration without reporting its failure rate under variable traffic; simulation studies typically evaluate a small number of configurations on a partial network. We evaluate a large scenario set on a complete network, and report the distribution of service degradation rather than a single optimum.

III. METHODOLOGY

A. Simulation Model

Fleet operations are modelled as a discrete-event system. Time advances from one event to the next rather than in fixed steps, which suits a system whose relevant events (arrival at a charging station, start and end of a charging session, departure on a trip) occur at irregular intervals. Each charging station is a resource with a finite number of connectors, a first-in-first-out queue, and a service time equal to the charging duration.

Buses depart on their assigned corridor according to the agency's target headway. State of charge (SoC), the percentage of battery energy remaining relative to full capacity [11], is decremented per inter-stop segment using the energy consumed on that segment:

$$SoC_i = SoC_{i-1} - \frac{EC(v_i) \times d_i}{E_{bat}} \times 100\% \quad (1)$$

where $EC(v_i)$ is energy consumption per kilometre at segment speed v_i , d_i is segment distance, and E_{bat} is battery capacity. When SoC falls to 30%, the bus leaves its route

for the nearest assigned station, queues if all connectors are occupied, charges, and resumes.

$EC(v)$ follows the rational polynomial form of Mamarikas et al. [9]:

$$EC(v) = \frac{av^2 + bv + c + \frac{d}{v}}{ev^2 + fv + g} \quad (2)$$

where a through g are vehicle-specific calibration coefficients. We chose this model because it is non-linear in traffic speed. When a bus moves slowly in congestion, the d/v term grows large, reflecting the steady draw of auxiliary systems like air conditioning over a longer time. This non-linearity matters because the stochastic layer described below perturbs speed. A linear model would average those perturbations away, but a non-linear one correctly drains the battery faster in variable traffic.

B. Stochastic Layer

A deterministic model uses one fixed value per input parameter and returns the same result on every run; a stochastic model treats uncertain parameters as random variables and must be repeated to characterise the distribution of outcomes. We adopt the latter, following Law [12].

Travel speed is drawn from a truncated log-normal distribution at two levels. A daily modifier with a 15% coefficient of variation applies to all corridors on the same simulated day, representing city-wide congestion that affects many routes at once. A per-segment residual with a 5% coefficient of variation adds local variation on top. Each scenario is run for $N = 100$ simulated days, and results are reported as mean and standard deviation across those days. Convergence testing confirmed that the headway mean stabilises well before 100 repetitions.

C. Evaluation Metrics

Two metrics drive the comparison. Mean realised headway $\bar{h}_r$ for route r is computed from consecutive departure times and compared against the agency's target h_r^{target} :

$$\bar{h}_r = \frac{1}{N_r - 1} \sum_{i=2}^{N_r} (t_{i,r} - t_{i-1,r}), \quad \Delta h_r = \bar{h}_r - h_r^{target} \quad (3)$$

The second metric counts *degraded routes* per day: routes whose service collapses because buses are stranded or their trips cancelled while waiting for a connector. Headway measures service quality; degraded routes measure service survival. A configuration can perform well on one and poorly on the other, and the distinction turns out to separate the two intervention strategies.

IV. CASE STUDY AND EXPERIMENTAL SETUP

The network comprises 20 corridors with published headway targets ranging from 10 to 45 minutes, operating from 05:30 to 20:30. Route geometry, stop sequences, and inter-stop distances are taken from the operator's network data.

Candidate station locations were derived from the simulation itself rather than assumed. We ran the fully electrified

network with charging demand unconstrained and recorded where buses naturally crossed the 30% SoC threshold, yielding 21,374 threshold events across the corridor network. The five stops with the highest event counts were taken as candidates. Four of them (Ngabean, Condong Catur, Kridosono, Adisucipto) collectively lie on all 20 corridors; the fifth, Janti Utara, serves four corridors that the other four already cover, so adding it as a fifth site contributes no marginal route coverage. The four-site set was therefore fixed for the capacity and policy experiments.

The model was calibrated qualitatively against the operator’s live baseline: two buses on corridor L-1 charging at Adisucipto. The simulation requires three charging sessions per day for that configuration where the real operation requires two. The deviation is consistent with the energy model’s origin, since Mamarikas et al. calibrated on heavy-duty vehicles heavier than the TransJogja units. We record this as a documented accuracy limit rather than correcting it with a fitted scale factor, which would trade a known bias for an unverifiable one. The direction and order of magnitude agree, which is the level of agreement the available field data supports.

V. RESULTS

A. Station Count Does Not Substitute for Allocation

Table I compares configurations holding total connector count constant while varying how many sites those connectors are spread across.

TABLE I
EFFECT OF STATION COUNT AND ALLOCATION AT EQUAL TOTAL CAPACITY

ID	Configuration	Headway (min)	Degraded/day
S15	1 site (20 connectors)	32.16 ± 2.21	2.47
S16	2 sites	35.55 ± 2.06	6.97
S17	4 sites, equal split	40.84 ± 2.19	11.44
S17b	4 sites, proportional	34.93 ± 1.81	3.03

Spreading capacity across more sites made service worse, not better. Moving from one site to two raised degraded routes from 2.47 to 6.97 per day, and splitting evenly across four sites nearly doubled that again. The mechanism is that an even split strands connectors at sites whose corridors carry few buses while starving the site serving the busiest corridors. Allocating the same 20 connectors in proportion to each site’s actual bus load (S17b) recovers most of the loss. Site count is not the decision variable; the match between per-site capacity and per-site demand is.

B. Capacity Exhibits a Regime Shift

We then varied total capacity at a single site to locate the boundary between a system that works and one that collapses (Table II and Fig. 1).

Degradation does not rise smoothly as capacity falls. Between 19 and 18 connectors it jumps from 5.52 to 8.85 routes per day, and the composition of the queue changes with it: above that point the queue is dominated by buses that wait

TABLE II
CAPACITY SWEEP AT A SINGLE STATION

ID	Connectors	Headway (min)	Degraded/day
S15	20	32.16 ± 2.21	2.47
S24	19	33.22 ± 2.93	5.52
S23	18	36.71 ± 2.78	8.85
S21	16	44.78 ± 4.68	12.73
S22	12	70.88 ± 9.12	15.45
S19	8	—	19.90
S20	4	—	20.00

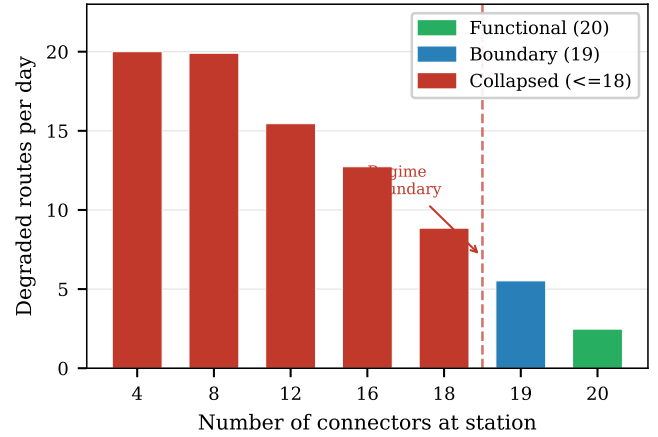


Fig. 1. Degraded routes per day as a function of connector count at a single station. The dashed line marks the regime boundary between 19 and 18 connectors, where removing one connector nearly doubles daily route degradation (5.52 to 8.85).

and eventually charge, below it by buses whose requests are cancelled and which end up stranded. Headway figures for the two smallest configurations are omitted because with 8 or fewer connectors essentially every route has collapsed, leaving too few completed trips for the headway statistic to mean anything.

The practical reading is that capacity decisions near this boundary are not marginal. A planner trimming one connector to save cost may be trimming the connector that separates a functioning network from a failing one, and the cost of being one unit short is not one unit’s worth of service.

C. Hardware and Scheduling Optimize Different Metrics

Table III and Fig. 2–3 compare the two intervention families against the proportional four-site baseline. The infrastructure family adds connectors above the baseline allocation; the algorithm family keeps baseline hardware and changes when buses decide to charge. V1 is time-aware, deferring charging based on remaining service time; V2 is queue-aware, additionally consulting current station occupancy.

Adding hardware buys headway. Raising capacity 12% above the proportional baseline (S28) cuts mean headway from 34.93 to 26.14 minutes, an 8.8-minute improvement, and reduces degradation to 0.01 routes per day. The marginal return is steep and then flat: the first step to +7% removes two thirds

TABLE III
INFRASTRUCTURE VERSUS ALGORITHM ($N = 100$ DAYS)

Scenario	Headway (min)	Degr./day	On target
S17b baseline	34.93 ± 1.81	3.03	0/20
S30 (+7% hw)	27.62 ± 1.57	1.01	4/20
S28 (+12% hw)	26.14 ± 1.52	0.01	4/20
S31 (V1 policy)	33.10 ± 1.91	0.01	0/20
S32 (V2 policy)	30.52 ± 2.12	0.74	0/20
S33 (S28 + V1)	30.78 ± 1.98	0.00	0/20

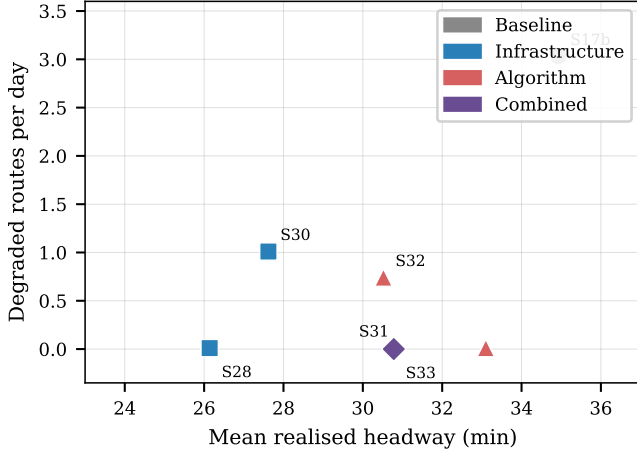


Fig. 2. Trade-off between mean realised headway and degraded routes per day for each scenario. Infrastructure scenarios (blue squares) improve headway; algorithm scenarios (red triangles) reduce degradation. Only the combined S33 (purple diamond) achieves zero degradation.

of the degradation, the next single connector removes almost all of what remains, and there is little left below that to justify further capacity.

The dynamic policy buys reliability instead. V1 reaches the same 0.01 degradation with no additional hardware at all, but its headway stays at 33.10 minutes, barely better than baseline, because deferring charging keeps buses in service at the cost of fragmenting their charging sessions. V2, which additionally reacts to queue length, performs worse than V1 on both metrics. Letting many buses observe the same queue state and defer simultaneously synchronises their decisions, so they exhaust their batteries together on the following trip; reacting to a shared signal turned out to be worse than not reacting at all.

Combining both (S33) produces the only configuration with zero degraded routes, and it holds that across a 14-day consecutive stress test in which battery state carries over between days. Its headway, 30.78 minutes, sits between the two single-lever options.

D. No Configuration Meets Published Targets

The rightmost column of Table III is the result the operator is likely to care about most: under a strict interpretation, where a route passes only if realised headway meets its published target exactly, the best configuration passes 4 of 20 routes.

Relaxing the criterion to a ± 10 -minute arrival tolerance, S28 passes 17 of 20.

The routes that pass share a property. All four (targets of 25, 25, 37 and 45 minutes) have loose published targets, and every route with a target below 25 minutes fails without exception; across the 19 routes excluding one extreme outlier, the correlation between target headway and headway deviation is strong and negative ($r = -0.845$). The added queueing and charging time is roughly uniform in absolute minutes across routes, so it consumes a much larger share of a 12-minute target than of a 45-minute one.

The outlier is a single route whose deviation exceeds triple the next worst. Its cause is micro-locational rather than systemic: it shares the Adisucipto station with a corridor running 12 buses while itself running only 3, so its buses queue behind a much larger fleet at the one station they can reach. This is a site-level assignment problem, not a network-level capacity problem, and it would not be fixed by adding capacity uniformly.

VI. DISCUSSION

The three results point in a consistent direction: for this network, the binding constraint is the match between capacity and demand at specific sites, not the aggregate quantity of either.

For agencies planning similar rollouts, the ordering of decisions matters more than the size of the budget. Allocating capacity in proportion to per-site bus load recovers more service than adding sites does. Capacity should then be sized with explicit awareness of the regime boundary, because the cost curve around it is asymmetric: being one connector over is cheap, being one connector under is not.

The infrastructure-versus-algorithm comparison does not resolve to a single winner because the two levers answer different questions. An agency whose priority is that no route collapses can reach that with software alone, accepting headway roughly at today's level. An agency whose priority is service regularity has to buy hardware; no policy we tested reached comparable headway without it. The combination achieves both but requires both investments, and our stress testing covered only the combined configuration, so the durability of the hardware-only option over consecutive days remains unverified.

A separate finding concerns what simulation can and cannot distinguish. A mixed charging mechanism, adding depot charging after the service window on top of opportunity charging during it, produced results identical to opportunity charging alone to two decimal places on every metric reported here. This is a consequence of the evaluation window rather than a property of the mechanism: depot charging after service ends affects only the next day's starting state, which a single-day evaluation cannot observe. Multi-day evaluation would be required to separate them, and we note this to prevent the null result being read as evidence that the mechanism does not matter.

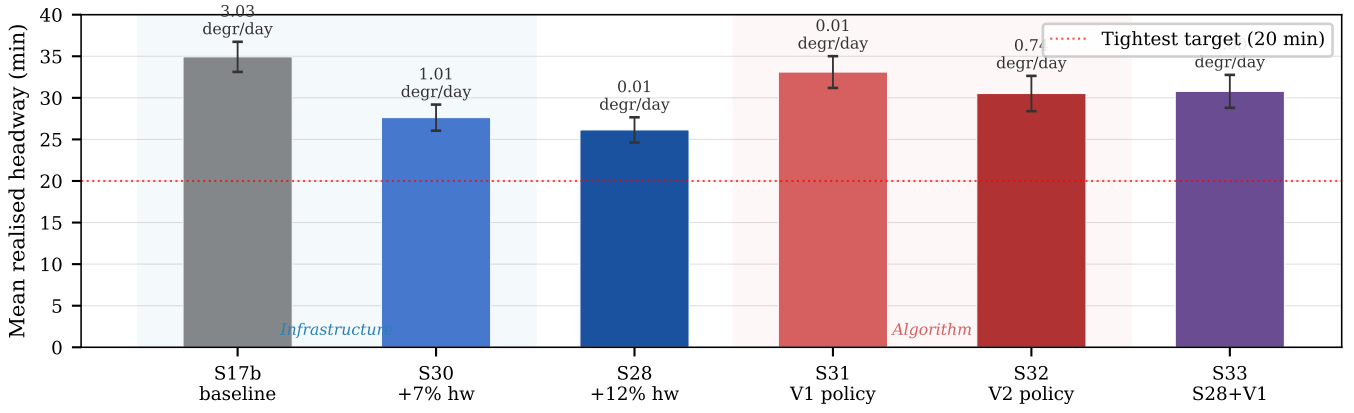


Fig. 3. Mean realised headway (with $\pm 1\sigma$ error bars) and degraded-route rate for each intervention scenario. The red dotted line marks the tightest agency target (20 min). Degraded-route values are annotated above each bar. Infrastructure scenarios (blue) achieve the best headway; algorithm scenarios (red) achieve the lowest degradation; the combined scenario S33 (purple) achieves zero degradation.

The most consequential result for policy may be the one that no configuration achieved. Since even the best scenario meets the published targets on a minority of routes, and since failure correlates with target tightness rather than with any charging decision, the gap is unlikely to close through charging investment alone. Either the targets were set for diesel operations and need revisiting for an electrified fleet, or closing the gap requires fleet expansion beyond the range examined here. Our results support stating that choice explicitly rather than treating the targets as fixed.

Several limitations bound these conclusions. The energy model overestimates consumption for this fleet, as the calibration showed, which makes the capacity requirements reported here conservative. Battery degradation over vehicle lifetime is not modelled. Passenger loading is held constant rather than varying with demand, so crowding effects on consumption are absent. Candidate sites were restricted to existing stops, so the results describe the best configuration within the operator's current network rather than a greenfield optimum.

VII. CONCLUSION

We evaluated charging infrastructure and charging policy for full electrification of a 20-corridor BRT network using discrete-event simulation with a Monte Carlo layer over 100 stochastic days and 31 scenarios. Distributing connectors across more sites degraded service when the split ignored per-site demand, while proportional allocation of the same capacity recovered it. Capacity showed a regime shift near 19 connectors, where removing one connector nearly doubled daily route degradation. Added hardware and dynamic scheduling improved different metrics, headway and degradation respectively, and only their combination held zero degraded routes through a 14-day consecutive stress test. No configuration met the operator's published headway targets on every route, and failure tracked how tight each target was rather than any charging decision, which places the remaining gap outside the scope of charging investment.

Future work should extend the multi-day stress test to the hardware-only configuration, whose durability over consecutive days remains untested, and evaluate mixed charging over a multi-day window where its effect can appear. Site-level assignment for corridors that share a station with much larger fleets also merits separate treatment, since the one extreme outlier in our results arises from that configuration rather than from aggregate capacity.

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